

1. Emotion Graph Generation

A   “Wow, that's amazing news!”

P_{in} Task: Recognize the emotion with keywords in English:
(A) Neutral (B) Happy (C) Sad (D) Surprised (E) Angry

For an audio and its associated question, generate an emotion graph in JSON format with these elements:

1. Acoustic features: pitch, speech rate, and volume.
2. Transcribed text, its sentiment, and textual keyword
3. Relationships between acoustic features and textual sentiment, indicating whether they support or conflict emotional expression.

E_{in}

Emotion Graph

$D_{audio} = “[A]”$

DSP

$P_{text} = “[P_{in}][E_{in}]”$

LLM

2. Response Generation

A   “Wow, that's amazing news!”

Emotion Graph:

E_g

```
{
  "acoustic":
    [{"pitch": "high"}, {"rate": "slow"}, {"volume": "loud"}],
  "text":
    [{"transcription": "Wow, that's amazing news!",
      "sentiment": "positive", "key word": "amazing"}],
  "relationships":
    [{"pitch": {"supports": ["sentiment"]},
      "volume": {"supports": ["sentiment"]}},
     {"rate": {"conflicts": ["sentiment"]}]}
}
```

C

Use the audio and emotion graph as context and answer the following question:

P_{in}

Task: Recognize the emotion with keywords in English:
(A) Happy (B) Neutral (C) Sad (D) Surprised (E) Angry

S

Select the option letter from the provided choices to answer

$P_{out} = “[A][E_g][C][P_{in}][S]”$



LALM

Response

$R = “B”$